

Problem 7.19

What is the value 8 years from now of \$500 today if $v = 0.96$?

Solution

Since

$$v = \frac{1}{1+i},$$

we have

$$1+i = \frac{1}{v} = \frac{1}{0.96},$$

so the accumulated value in 8 years of \$500 is

$$500(1+i)^8 = 500 \left(\frac{1}{0.96} \right)^8.$$

Equivalently, since v^8 is the discount factor for 8 years,

$$\text{Value at time 8} = \frac{500}{v^8} = \frac{500}{0.96^8}.$$

Computing this,

$$\frac{500}{0.96^8} \approx 693.1067.$$

Therefore, the value 8 years from now is

$$\boxed{\$693.11}.$$